Section 8 | Responsible Technology Ethics and Impact

CIS480 Capstone • T10 team evidence package • November 1, 2026

Project: Predictive maintenance classification using the public synthetic AI4I 2020 dataset. Team: Keeshaun Cobb (report and integration), Brett Harrison (modeling and safeguards), Wynn Nguyen (impact and risk analysis), Zach Lee (visualizations).

8.1 PURPOSE AND DECISION

The project compares Random Forest and Logistic Regression for identifying likely equipment failures from operational measurements. Its intended value is to support earlier maintenance review, not to automate safety-critical decisions. For this academic prototype, the team recommends the cleaned-data Random Forest as the stronger candidate for further validation, conditional on a human-reviewed workflow. It is not approved for live deployment.

The Week 10 decision was to implement explicit model-performance checks, document harms and mitigations, and restrict the use of predictions to decision support. The evaluation rules are provisional project criteria: recall ≥ 75% and false-positive rate (FPR) ≤ 2%. Passing both is necessary for further consideration, but not evidence of field readiness.

8.2 IMPACT EVALUATION

Wynn's submitted impact evaluation identifies the tradeoff: better failure detection may reduce unplanned downtime, while false alarms can create unnecessary inspections. Missed failures could cause downtime or equipment damage; excessive alarms can burden technicians and reduce trust. Since AI4I 2020 is synthetic, field performance under real operating conditions is unknown. The team therefore requires human review, transparent metric reporting, and eventual testing on representative authorized equipment data before any operational pilot.

Affected parties include maintenance technicians (inspection workload), operators (downtime and disruptions), plant managers (resource allocation), and safety personnel (consequences of missed failures). The model must not be presented as guaranteeing equipment safety.

8.3 RISK REGISTER

Risk	Impact	Likelihood	Potential consequence	Mitigation / control
R1 Missed failures	High	Medium	Undetected failures may lead to downtime or damage.	Monitor recall; manual escalation for critical symptoms; validate on field data.
R2 False alarms	Medium	High	Unnecessary inspections and alarm fatigue.	Track FPR and alert volume; tune threshold

				after field review.
R3 Synthetic-to-real gap	High	Medium	Real equipment may differ from simulated patterns.	Prohibit production use pending representative field validation.
R4 Unreliable predictions	High	Medium	Model quality may vary with different data.	Retest on new data; monitor drift; retain human override.
R5 Misinterpretation of visuals	Medium	Medium	Users may mistake class accuracy or charts for proof of safety.	Label synthetic context, denominator, and limitations.

Impact and likelihood are qualitative team judgments, not measured field incident rates. R1–R4 adapt Wynn's submitted register; R5 extends it to dashboard communication.

8.4 IMPLEMENTED MITIGATION AND INSPECTABLE EVIDENCE

Brett committed "Minimum Performance Checks" cells to all four model notebooks in commit 0b41dca. Each computes or reads recall, computes FPR as FP/(FP+TN), and prints PASS/FAIL against recall ≥ 0.75 and FPR ≤ 0.02. The committed notebook outputs show both passing and failing recall cases, demonstrating that the rule can reject a model that misses the required sensitivity.

Inspectable source and dated change: https://github.com/37216110/cis480_capstone/commit/0b41dca295ce334ce91a7c78fcc448a9770dedee

Model / dataset	Recall check	FPR check	Decision under provisional rules
Random Forest / cleaned	78.79% — PASS	1.45% — PASS	Eligible for further evaluation
Random Forest / original	77.94% — PASS	1.92% — PASS	Eligible for further evaluation
Logistic Regression / cleaned	16.67% — FAIL	0.21% — PASS	Do not recommend
Logistic Regression / original	10.29% — FAIL	0.21% — PASS	Do not recommend

The saved outputs were inspected in the public commit; this package does not claim an independent full rerun. These notebook checks report results rather than enforce an automated deployment block. A future production gate must stop release when a criterion fails.

8.5 VALIDATION RECORD

The companion file T10_validation_record.csv records each question, expected result, observed result, status, and inspectable evidence. Key tests: cleaned Random Forest passes

both criteria; cleaned Logistic Regression fails recall despite low FPR. The negative case matters because high overall accuracy on imbalanced data would otherwise hide missed failures.

Cleaned Random Forest test denominator: 1,995 records, including 66 actual failures and 1,929 actual nonfailures. TP=52, FN=14, FP=28, TN=1,901. Recall=52/66=78.79%; FPR=28/1,929=1.45%. Precision=52/(52+28)=65.00%, so 28 of 80 predicted failures were false alarms. These values match the saved notebook results reported in prior team evidence.

8.6 INTENDED AND PROHIBITED USES

Intended: academic evaluation of failure-prediction methods; comparison of classification metrics; visualization of synthetic equipment patterns; and, after separate real-world validation, a possible human-reviewed maintenance triage aid.

Prohibited: automatic equipment shutdown, autonomous repair orders, safety certification, employee performance scoring, regulatory compliance determinations, or live production deployment on the basis of synthetic-data scores alone. No prediction should override a technician's inspection or existing safety procedures.

8.7 AI FEATURE DOCUMENTATION AND HUMAN OVERSIGHT

Item	Documentation
Model / service	Local scikit-learn Random Forest classifier; Logistic Regression baseline. No hosted generative AI service is part of inference.
Data used	Public synthetic AI4I 2020 operational measurements: air temperature, process temperature, rotational speed, torque, tool wear, and relevant encoded inputs; failure labels used for supervised training/evaluation, not as prediction inputs.
Data sent	Inputs are processed within the analysis notebooks. No external inference endpoint or private sponsor dataset is documented.
Human oversight	A qualified reviewer interprets the prediction, checks sensor and equipment context, and decides whether to inspect.
Failure behavior	If checks fail, data are missing/out of range, or the model cannot run, do not treat output as clearance; refer to established manual procedures.
Output verification	Use held-out confusion matrix, recall, precision, FPR and baseline comparison; review individual alerts and misses before any field use.

8.8 LIMITATIONS AND EVIDENCE BOUNDARIES

Supported: the committed notebooks contain threshold logic and saved pass/fail outputs; cleaned Random Forest meets both provisional criteria on its synthetic held-out sample;

Wynn documented the principal ethical risks. Not supported: performance on real machinery, a validated operational alert threshold, an automatic production-release block, field costs of missed failures or inspections, or sustained performance under drift. Original-versus-cleaned notebooks used different splits, so changes in scores cannot be attributed solely to removal of 27 inconsistent observations.

Zach reported that he corrected his visuals. The updated workbook is to be uploaded to the GitHub repository and its latest contents and accessibility to be independently verified.

8.9 CURRENT WORK PLAN AND CONTRIBUTIONS

Owner	Task	Status	Dependency	Acceptance criterion / contribution evidence
Brett	Add recall/FPR checks to notebooks	Committed; saved outputs inspected	Existing model notebooks	All four notebooks show threshold logic and results; commit 0b41dca
Wynn	Impact evaluation and risk register	Delivered	Model selection and synthetic-data context	Impact Evaluation.docx; four risks with mitigation
Zach	Correct Tableau visuals and limitations	Reported complete	Cleaned dataset / workbook	Updated .twb/.twbx opens, labels match model evidence; link/file pending
Keeshaun	Write Section 8, validation, uses, disclosures, package	Draft assembled; team review pending	All member evidence	Accessible report + CSVs + linked code; team confirms final version

8.10 SOURCES, REPRODUCIBILITY, AND AI-USE DISCLOSURE

Primary sources: Wynn, Impact Evaluation.docx (team-authored, supplied for T10); Brett Harrison, GitHub commit 0b41dca (model notebook changes and saved outputs); prior team model evaluation evidence; UCI AI4I 2020 Predictive Maintenance Dataset (public synthetic data).

AI-use disclosure: ChatGPT assisted with drafting and organizing Section 8, structuring the risk and validation tables, and clarifying limitations.